

Session 3: Deep Learning for Wafer Defect Detection using CNNs



VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Value Added Program (Hybrid)

VAC2514 Artificial Intelligence in Semiconductor Industry Applications



Prof Elakkiya R

Assistant Professor, Department of Computer Science &
Associate Head, APPCAIR, BITS Pilani Dubai Campus
Chair, ACM-W Professional Chapter –Dubai
Vice Chair, ACM Professional Chapter –Dubai
Stanford-Elsevier Top 2% World Scientists (2024)

Email: elakkiya@dubai.bits-pilani.ac.in

Expertise

- Artificial Intelligence, Deep Learning, Cognitive and Affective Computing and Multimodal Data Fusion
- Led 16 Projects (SERB, DST, DRDO-iDEX, Royal Society UK, AICTE, Trjaectorie, TVS Credits, TVS Motors, etc.)
- Authored 75+ papers, 6 patents, 6 books and 7 chapters

Profile:  |  |  |  |  | 

Overview



Why classical ML fails on wafer images



CNN Intuition



CNN Architecture Overview



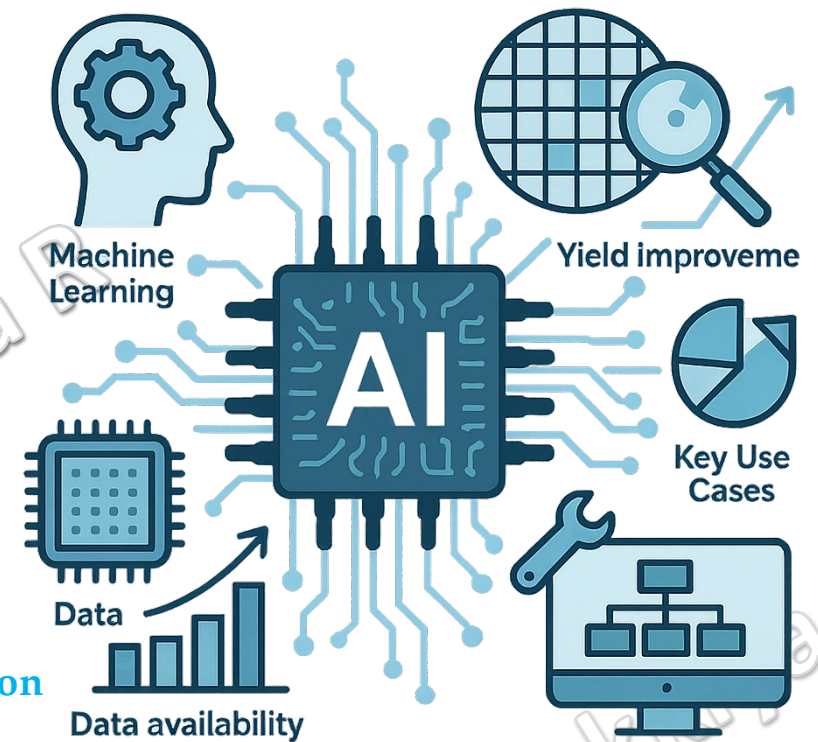
ML vs DL & when to use DNN



Why Transfer Learning Matters in Semiconductor Vision



Other Deep Learning Paradigms



Overview



Why classical ML fails on wafer images



CNN Intuition



CNN Architecture Overview



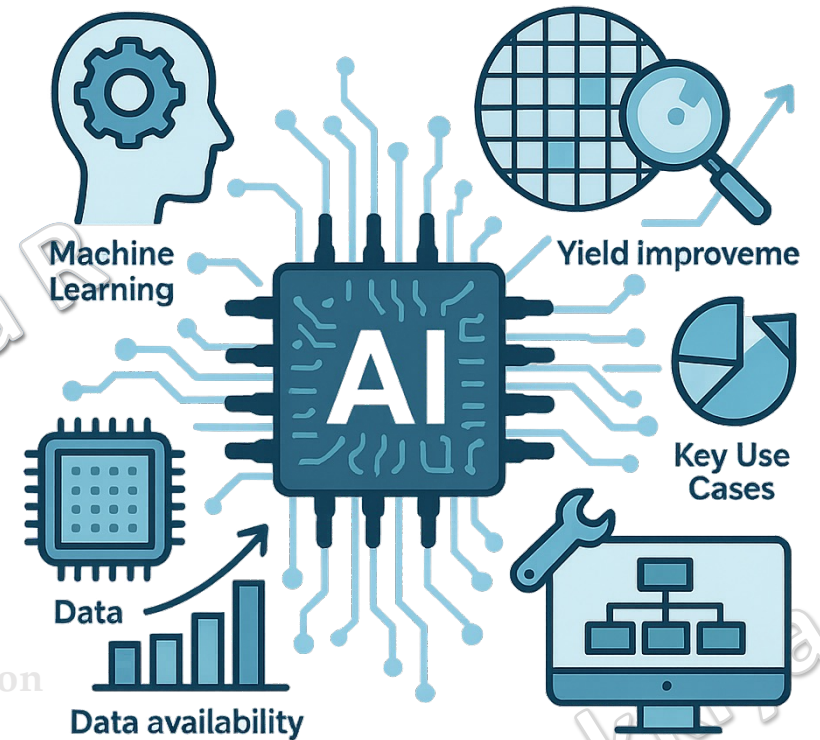
ML vs DL & when to use DNN



Why Transfer Learning Matters in Semiconductor Vision



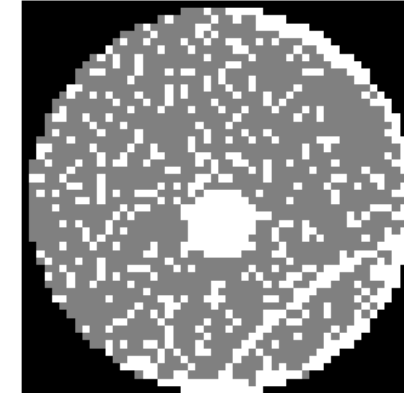
Other Deep Learning Paradigms



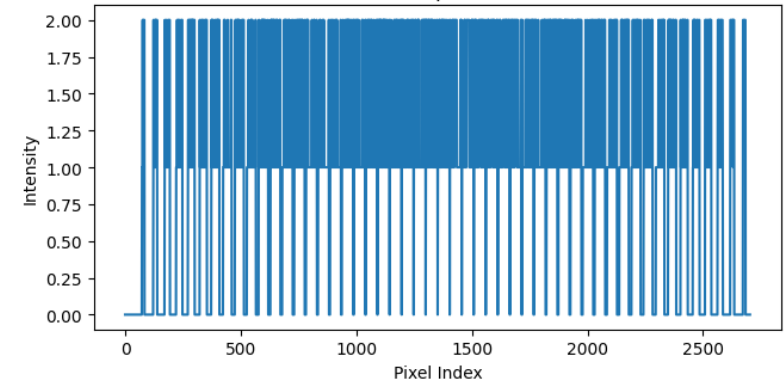
Why Classical ML Falls Short?

- Traditional ML models (e.g., SVM, Random Forests) rely on:
 - Manually extracted features (texture, shape descriptors)
 - Flattened image representations (losing spatial context)
- Wafer defects often exhibit:
 - Spatial patterns:** rings, blobs, scratches, edge shifts
 - Non-linear features** difficult to hand-craft
- Need to **preserve 2D structure** of images → **CNNs are ideal**

Raw Wafer Image (Label: [1 0 1 0 0 0 1 0])



Flattened Vector (Spatial Context Lost)



Why Deep Learning for Wafer Defect Detection?

Wafer defects are spatially encoded

- Ring, edge, blob, scratch patterns
- Classical ML loses spatial layout during flattening

CNNs preserve spatial locality

- Use filters that slide over image patches
- Detect local structures like edges and curves

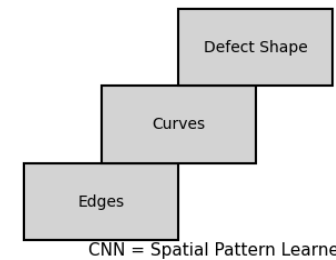
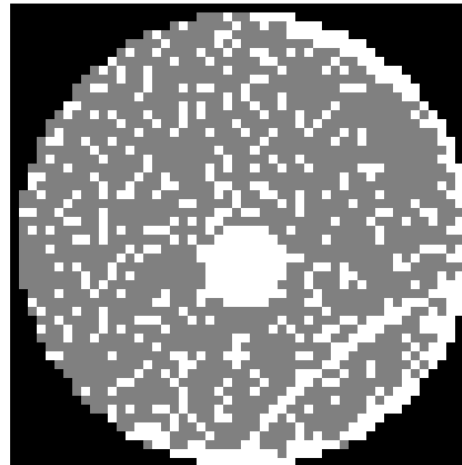
Feature learning is hierarchical

- Shallow layers → edges/blobs
- Deep layers → complex defect morphologies

Adaptable to process variations

- CNNs learn directly from images
- No need for handcrafted features

Raw Wafer Image (Label: [1 0 1 0 0 0 1 0])



Overview



Why classical ML fails on wafer images



CNN Intuition



CNN Architecture Overview



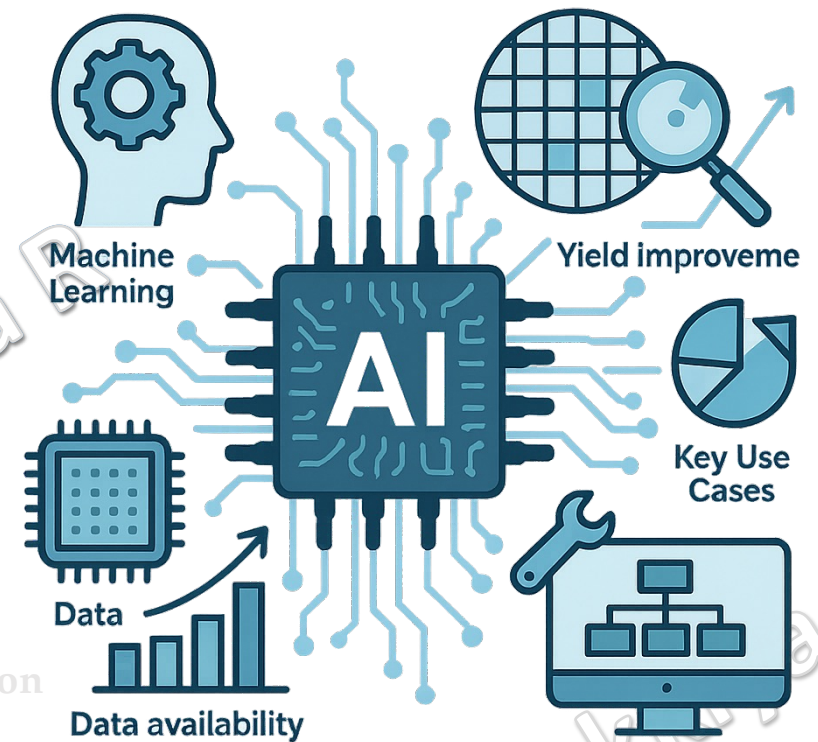
ML vs DL & when to use DNN



Why Transfer Learning Matters in Semiconductor Vision



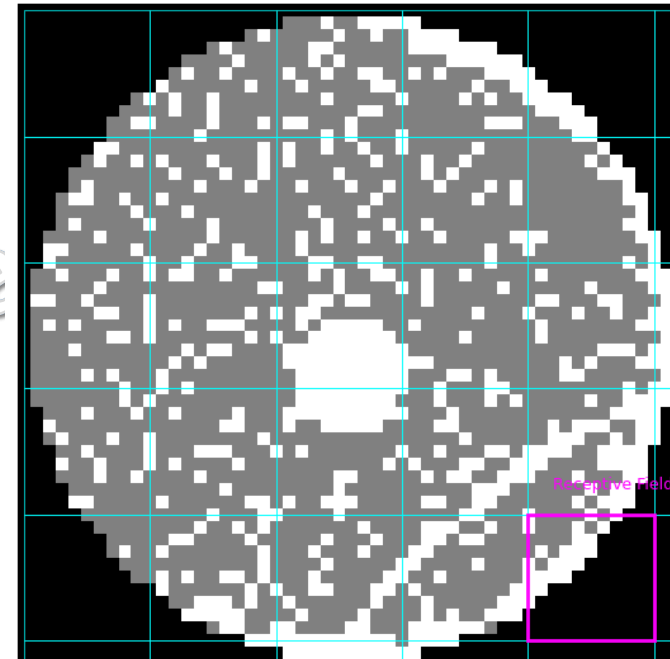
Other Deep Learning Paradigms



CNN Intuition (1)

- ◆ A **Convolutional Neural Network (CNN)** is a deep learning model specialized for **grid-like data** such as images.
- ◆ CNNs apply **small trainable filters** (also called **kernels**) to **local regions** of the input image — these regions are called **receptive fields**.
- ◆ The same filter slides across the image to detect **spatial patterns** (e.g., scratches, edges, rings).
- ◆ Each filter creates a **feature map** that highlights where its learned pattern appears in the image.

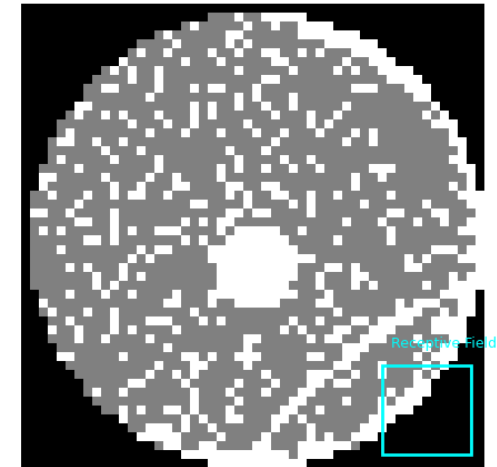
Wafer Image with Sliding Receptive Fields (Label: [1 0 1 0 0 0 1 0])



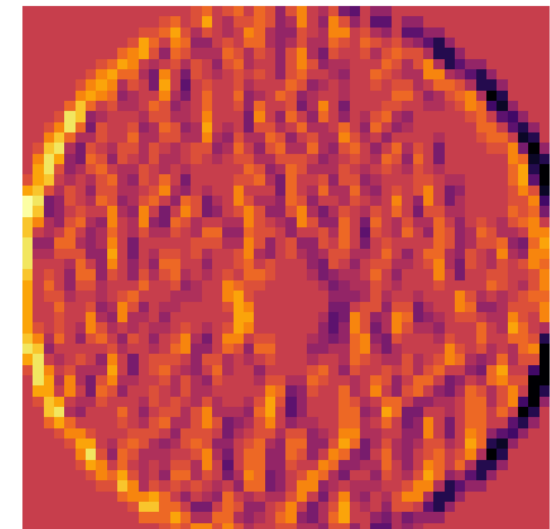
CNN Intuition (2)

- ◆ Each **filter operation** produces a **feature map** — a new representation of the image highlighting only what the filter cares about (like an edge or curve).
- ◆ CNNs use **multiple layers**:
 - ◆ **Shallow layers** learn edges and simple shapes.
 - ◆ **Middle layers** learn curves, blobs, and orientations.
 - ◆ **Deeper layers** detect complex **defect structures**.
- ◆ As you go deeper, the model captures **higher-level abstraction** — turning raw pixels into **semantic understanding**.
- ◆ This enables CNNs to **generalize** across wafers, lots, and process steps — even when noise or variation exists.

Input Wafer Image with Receptive Field



Feature Map after Edge Filter



Overview



Why classical ML fails on wafer images



CNN Intuition



CNN Architecture Overview



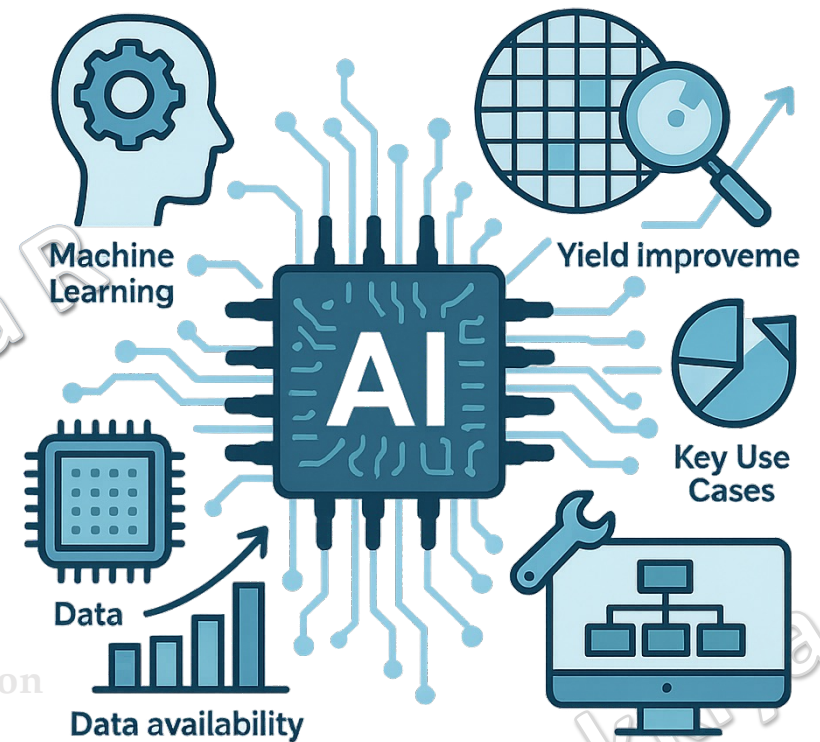
ML vs DL & when to use DNN



Why Transfer Learning Matters in Semiconductor Vision



Other Deep Learning Paradigms



CNN Architecture Overview

◆ A typical CNN processes an image through multiple layers:

- ◆ Convolutional Layer → Extracts local features using filters
- ◆ Activation Function (ReLU) → Introduces non-linearity
- ◆ Pooling Layer → Downsamples, retains dominant patterns
- ◆ Fully Connected Layer → Combines all features for classification

7	2	3	3	8
4	5	3	8	4
3	3	2	8	4
2	8	7	2	7
5	4	4	5	4

$$\begin{array}{|c|c|c|} \hline 1 & 0 & -1 \\ \hline 1 & 0 & -1 \\ \hline 1 & 0 & -1 \\ \hline \end{array} * \begin{array}{|c|c|c|} \hline 6 & & \\ \hline & & \\ \hline & & \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline 6 & & \\ \hline & & \\ \hline & & \\ \hline \end{array}$$

$7 \times 1 + 4 \times 1 + 3 \times 1 + 2 \times 0 + 5 \times 0 + 3 \times 0 + 3 \times -1 + 3 \times -1 + 2 \times -1 = 6$

◆ The model learns filters automatically via backpropagation.

◆ In wafer images:

- ◆ Early layers may detect edges and scratches
- ◆ Middle layers form shapes like rings or voids
- ◆ Final layers identify specific defect classes

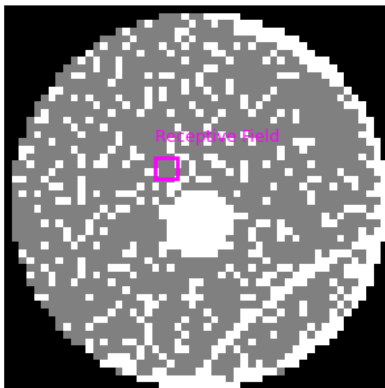
Convolutional Layer – The Feature Extractor

- ◆ Convolution applies a small filter (e.g., 3×3) over the image to detect **local patterns**.
- ◆ It computes a **dot product** between the filter and image patch (receptive field).
- ◆ Each filter generates a **feature map** showing where a pattern appears.
- ◆ Multiple filters → Multiple feature maps (edges, lines, blobs)

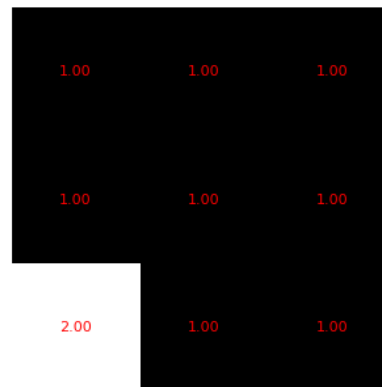
1	1	1	+1	0	-1
1	1	1	+1	0	-1
2	1	1	+1	0	-1

$$\begin{aligned}
 \text{Patch} \times \text{Kernel} &= (1 \times 1) + (1 \times 0) + (1 \times -1) + \\
 &\quad (1 \times 1) + (1 \times 0) + (1 \times -1) + (2 \times 1) + (1 \times 0) + (1 \times -1) \\
 &= 1 + 0 - 1 + 1 + 0 - 1 + 2 + 0 - 1 \\
 &= 1.00
 \end{aligned}$$

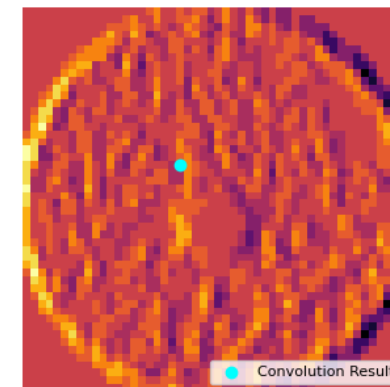
Wafer Image
(Label: [1 0 1 0 0 0 1 0])



3×3 Patch Extracted



Feature Map
(Patch → 1.00)



Activation Layer – Adding Non-Linearity (ReLU)

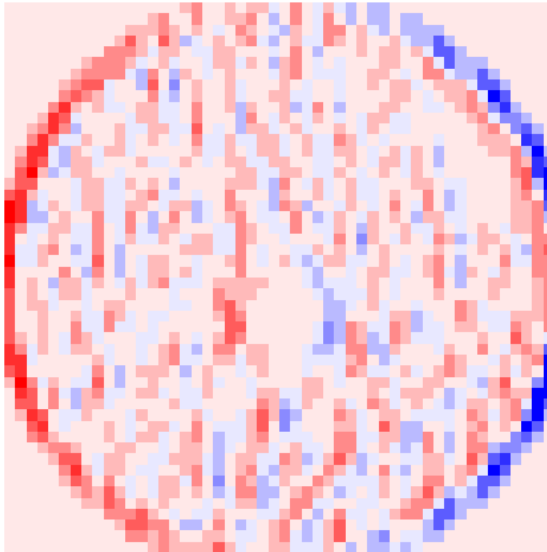
- ◆ ReLU (Rectified Linear Unit): $f(x) = \max(0, x)$
- ◆ Makes the CNN **non-linear** — essential for learning complex functions.
- ◆ Keeps **positive activations**, suppresses **noise** and weak signals.
- ◆ Applied element-wise to feature maps after convolution

Convolution Output = 1.00

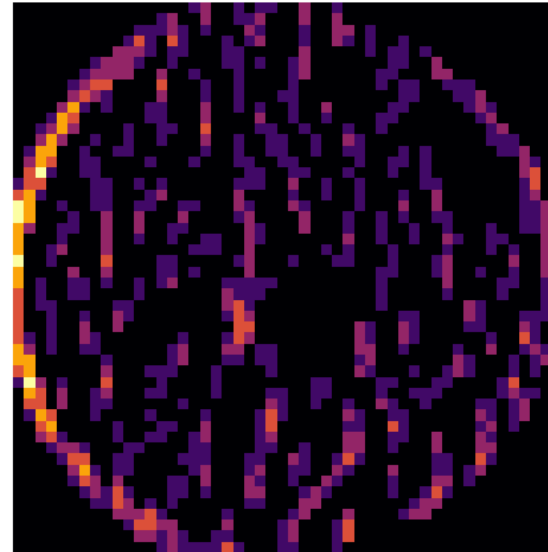
$\text{ReLU}(x) = \max(0, x) = \max(0, 1)$

ReLU Output = 1.00

Before ReLU (Feature Map with Negatives)



After ReLU (Only Positive Activations)



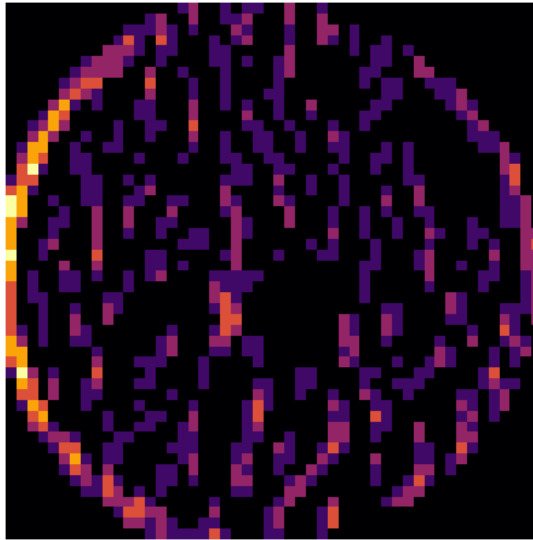
Pooling Layer – Downsampling Feature Maps

- ◆ Reduces spatial size while retaining **important features**
- ◆ Common method: **Max Pooling**
- ◆ Takes the max value in a small patch (e.g., 2×2)
- ◆ Makes model **faster, more robust to shifts**

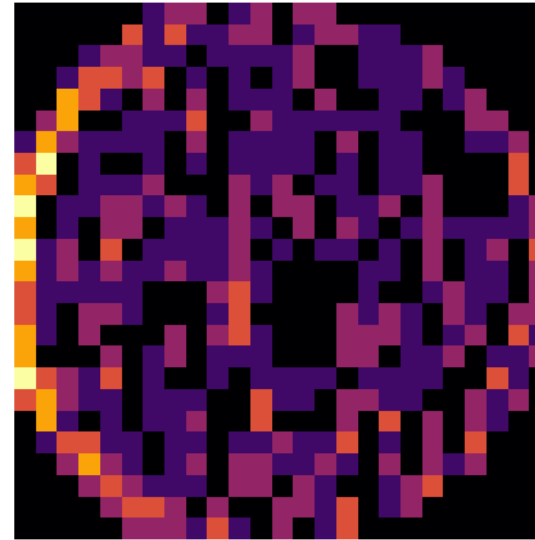
1	1
1	2

$$\text{Max Pool} = \max(1, 1, 1, 2) = 2$$

Before Pooling (ReLU Feature Map)



After 2×2 Max Pooling
(Downsampled Feature Map)



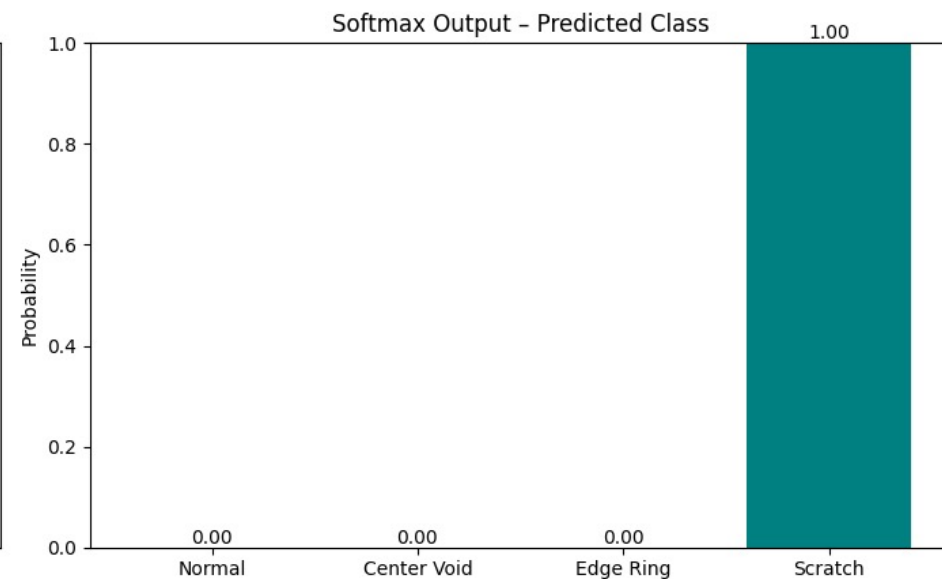
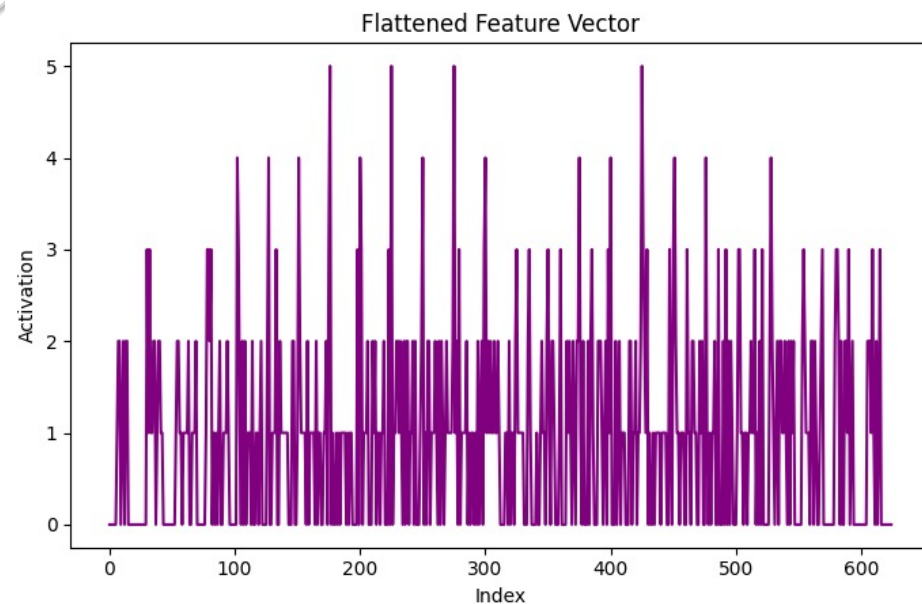
Fully Connected Layer – Making the Decision

- ☺ Takes all features learned from previous layers (flattened)
- ☺ Passes them into a standard neural network layer
- ☺ Learns **global patterns** and classifies the image

Logits: [2.25 1.95 1.45 2.6]

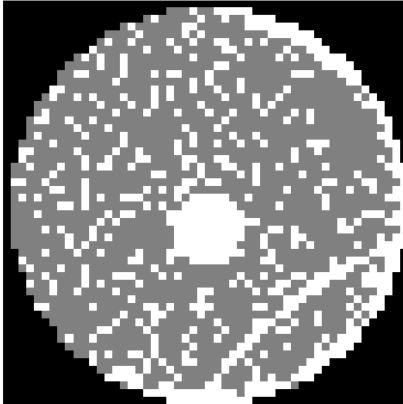
Softmax Probabilities: [0.2771 0.2053 0.1245 0.3932]

Predicted Class: 3

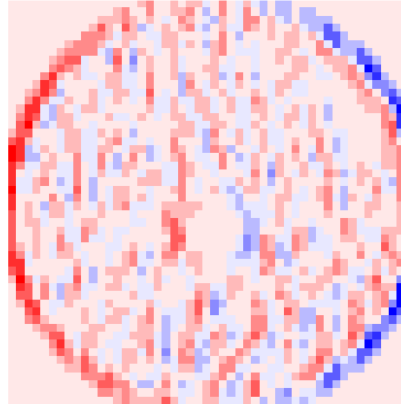


CNN Architecture Summary

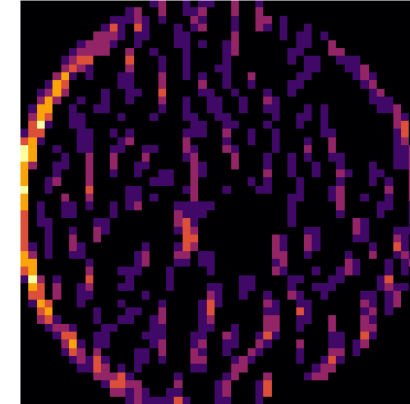
1. Input Wafer Image



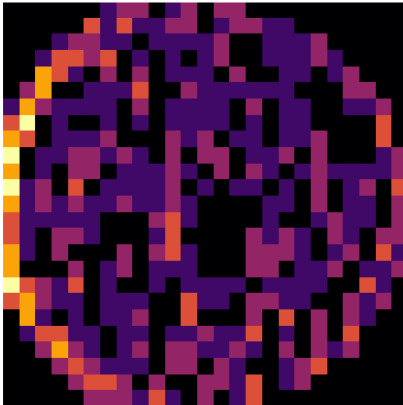
2. Feature Map (Conv)



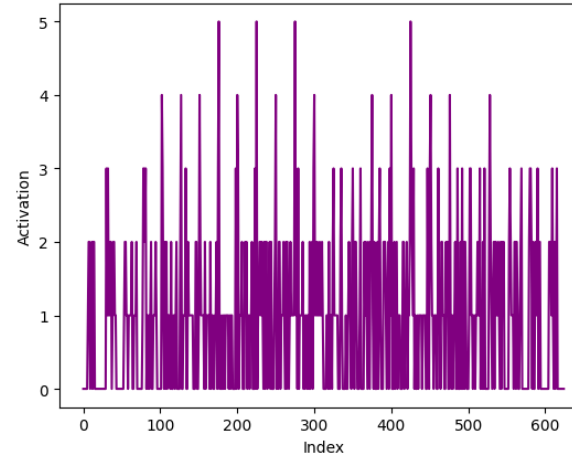
3. After ReLU



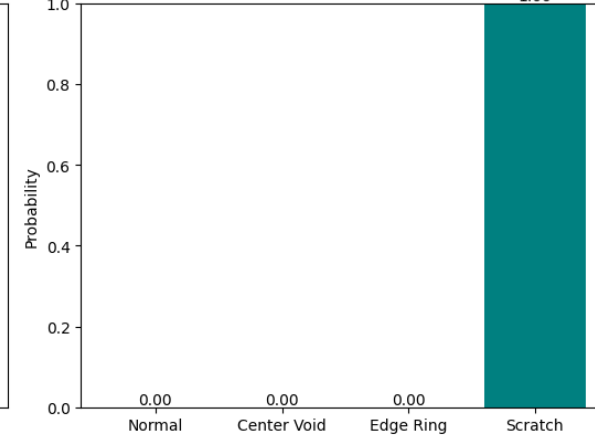
4. Max Pooled Map



5. Flattened Feature Vector

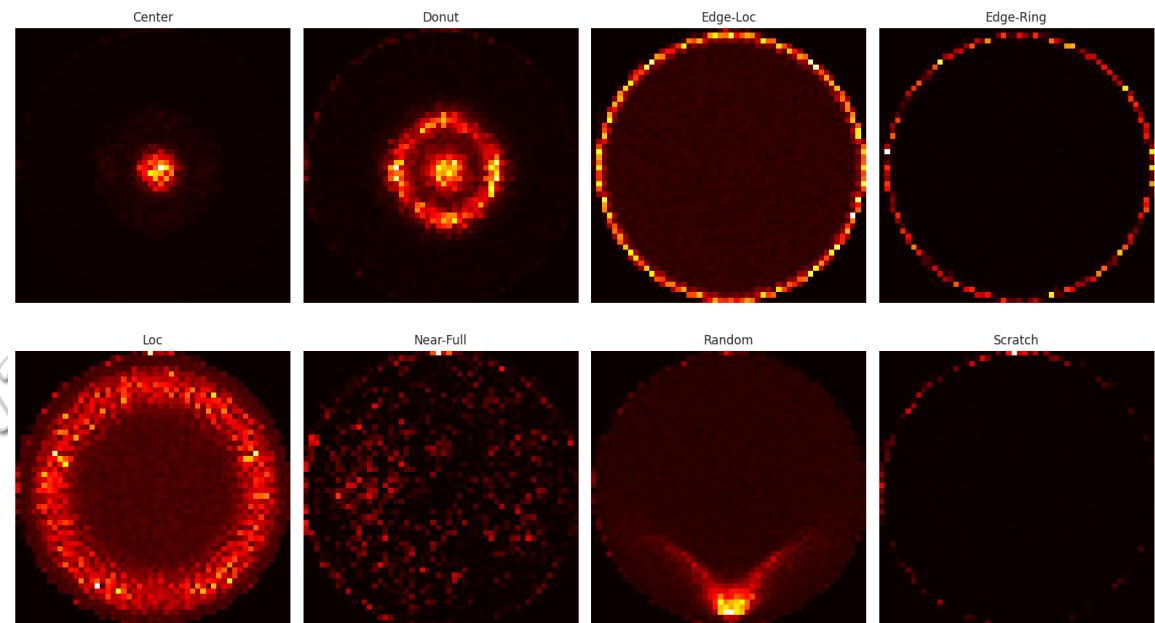


6. Softmax Classification



Visual Explainability with Grad-CAM

- ◆ CNNs are often called “black boxes” ?
- ◆ Grad-CAM = Gradient-weighted Class Activation Mapping
- ◆ Shows **which regions of the wafer image** most influenced the prediction
- ◆ Fab engineers use it to **visually validate**:
 - ◆ Was the defect location correctly identified?
 - ◆ Is the model focusing on noise or true patterns?



Overview



Why classical ML fails on wafer images



CNN Intuition



CNN Architecture Overview



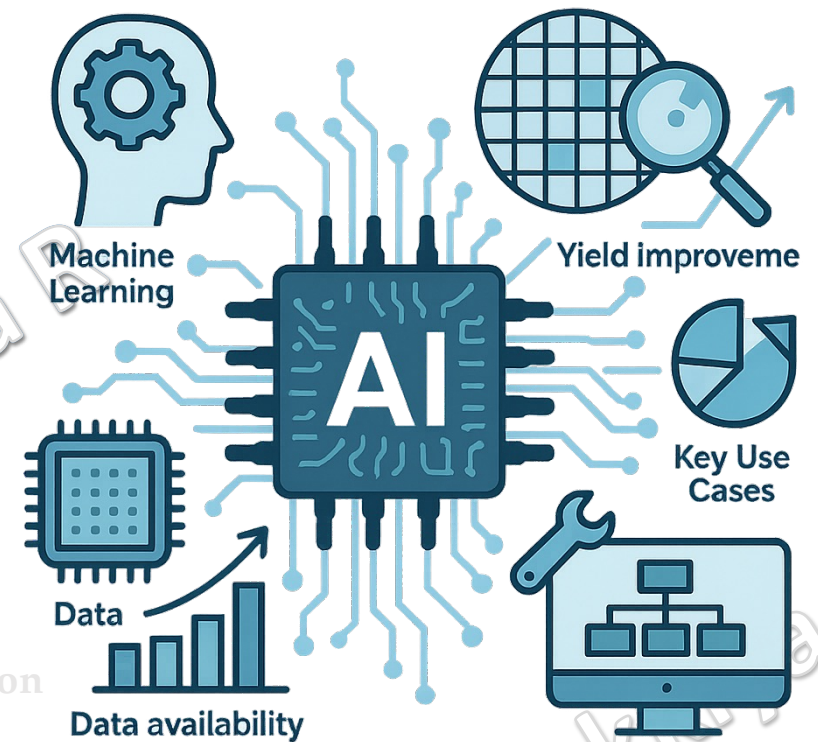
ML vs DL & when to use DNN



Why Transfer Learning Matters in Semiconductor Vision

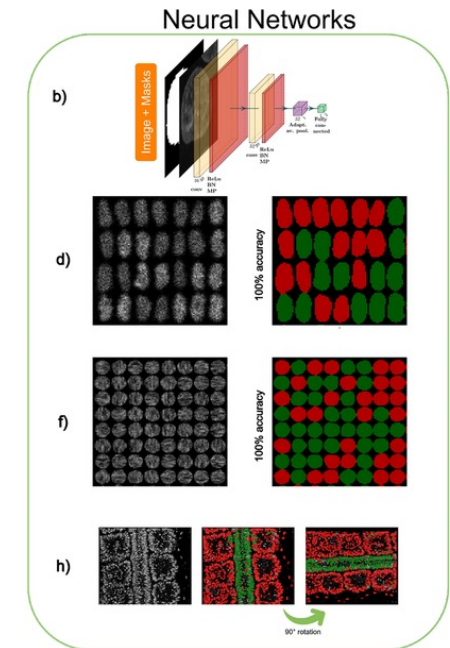
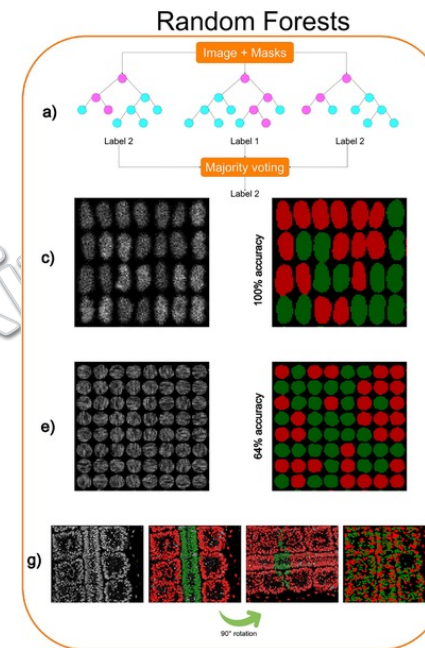


Other Deep Learning Paradigms



Why CNN > Traditional ML for Wafer Images

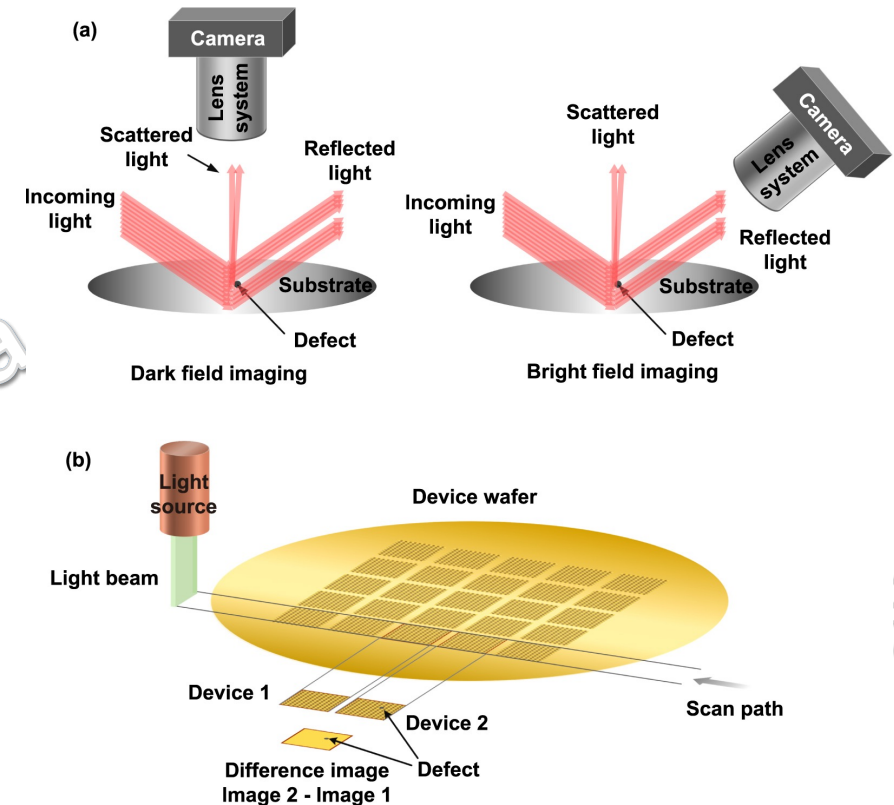
Feature	Random Forest (RF)	Convolutional Neural Net (CNN)
Input Type	Flattened vector	2D spatial image
Feature Engineering	Required manually	Learns features automatically
Spatial Pattern Capture	Poor	Excellent
Works on Raw Image	No	Yes
Generalization (Fab to Fab)	Limited	Strong with enough data
Explainability	Limited (feature import.)	Grad-CAM, attention maps



Src: Svetlana a supervised segmentation classifier for Napari

When to Use CNNs in Semiconductor Contexts

Application Area	Description
Optical Wafer Inspection	Detect surface defects (scratches, rings)
SEM Image Classification	Identify nanoscale issues (voids, bridges)
Root Cause Analysis (RCA)	Trace defect types to process steps
SPC Alerts (Statistical Process)	Image-based control triggers
Cross-Fab Transfer Learning	Generalize across tools/lines with tuning



Src: <https://iopscience.iop.org/article/10.1088/2631-7990/ac64d7>

Why Transfer Learning Matters in Semiconductor Vision

Core Idea:

- ◆ Training deep models from scratch = high data, time, and compute demand.
- ◆ Wafer datasets often limited per defect type.
- ◆ Transfer learning leverages pre-trained models (e.g., on ImageNet) and adapts them to wafer images.

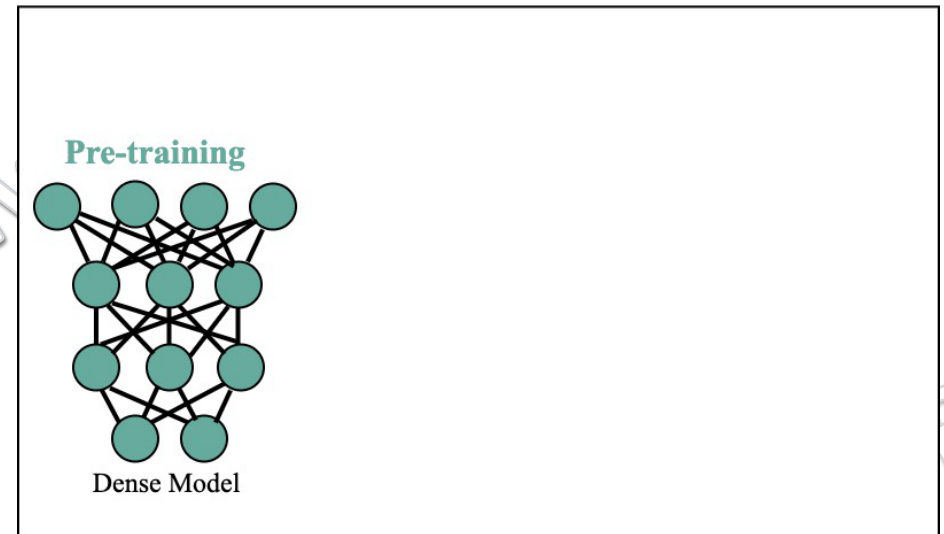


Image Src: <https://ai.plainenglish.io/the-revolution-of-pre-trained-nlp-models-unleashing-the-power-of-language-c2a8c4faf068>

Overview



Why classical ML fails on wafer images



CNN Intuition



CNN Architecture Overview



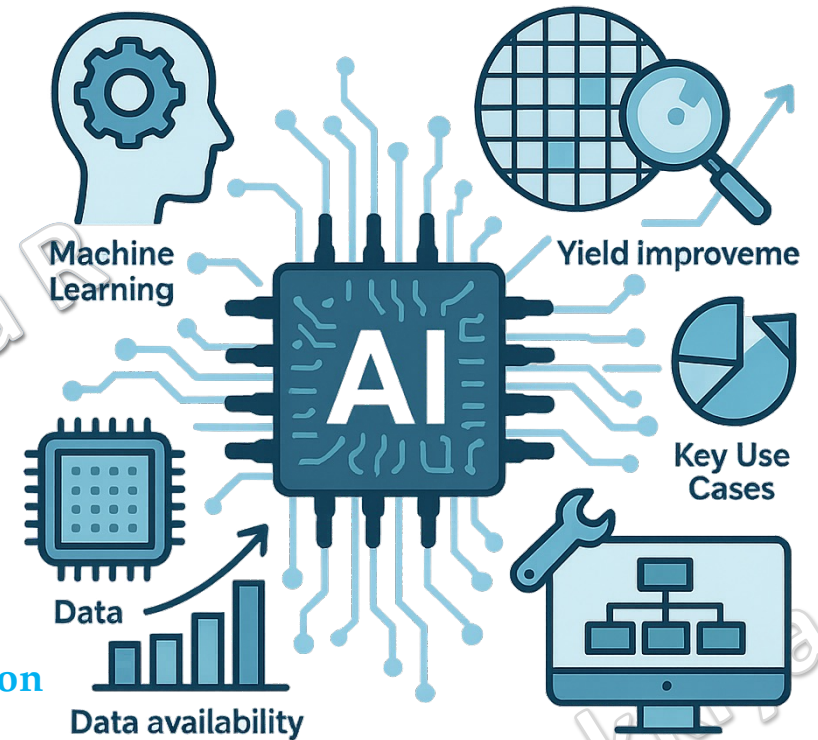
ML vs DL & when to use DNN



Why Transfer Learning Matters in Semiconductor Vision

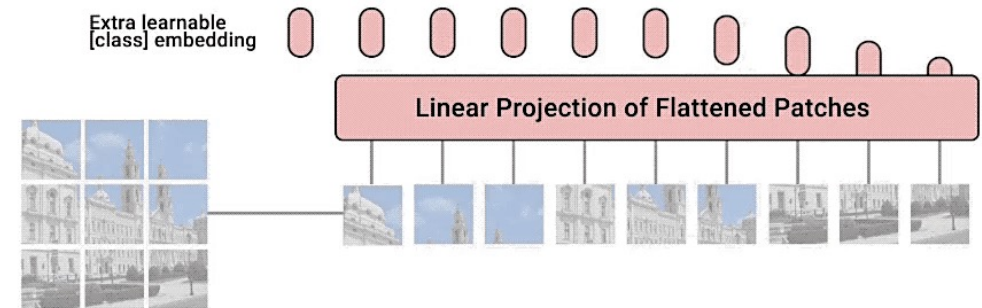


Other Deep Learning Paradigms



Popular Pretrained Models for Wafer Classification

Model	Strengths	Notes
ResNet50	Deep, residual connections for stability	Good general baseline
MobileNetV2	Lightweight, fast inference	Suitable for fab deployment
EfficientNet	Balanced accuracy and efficiency	Good for high-performance use
VGG16	Simple, interpretable features	Good for teaching / demo
DenseNet121	Strong feature reuse via dense connections	High accuracy, but memory-heavy
InceptionV3	Multi-scale feature detection	Good for complex, multi-defect images
Vision Transformer (ViT)	Global attention, learns long-range interactions	Needs larger wafer datasets
ConvNeXt-T	CNNs with transformer-like performance	Powerful and recent architecture
NASNet	Auto-searched architecture	May be overkill for small datasets



Overview



Why classical ML fails on wafer images



CNN Intuition



CNN Architecture Overview



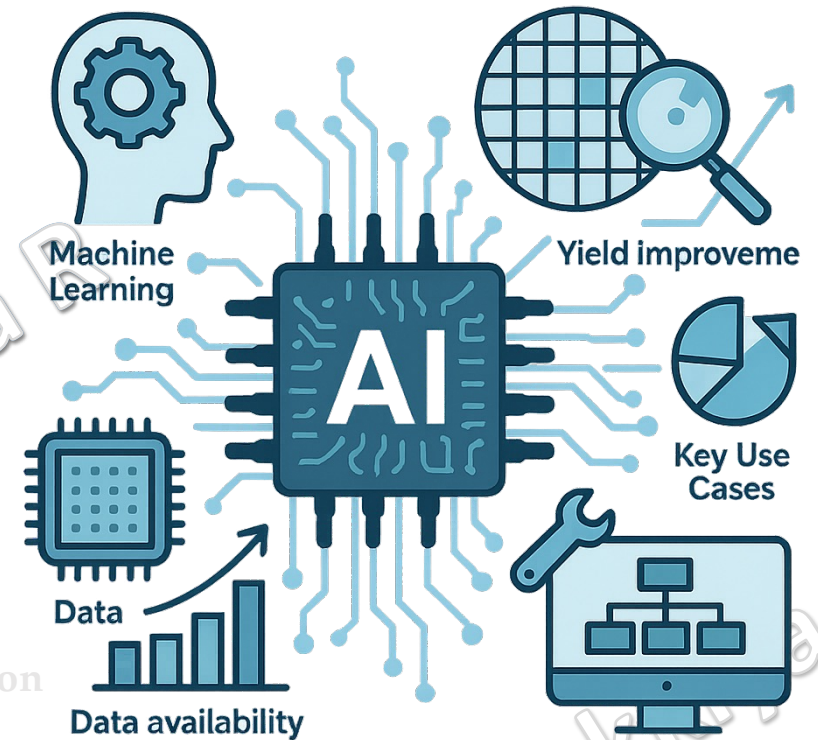
ML vs DL & when to use DNN



Why Transfer Learning Matters in Semiconductor Vision



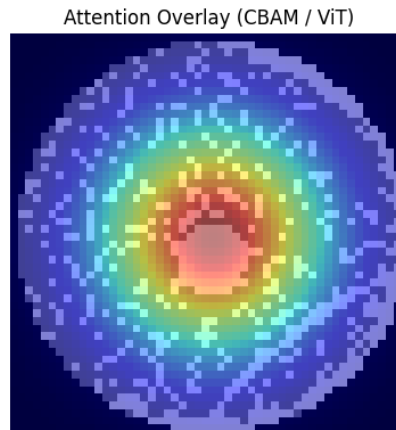
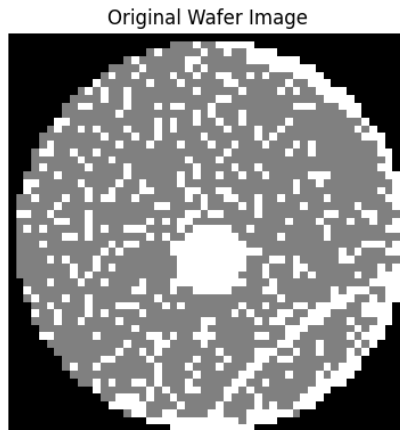
Other Deep Learning Paradigms



Beyond CNNs – Attention and Transformers in Wafer Analysis

Key Concepts:

- ◆ CNNs treat all spatial regions equally.
- ◆ But in wafer images, only *small regions* carry diagnostic signals (e.g., void, scratch).
- ◆ **Attention mechanisms** help models focus on the most relevant parts of the image.



Methods:

◆ CBAM (Convolutional Block Attention Module):

- ◆ Adds spatial + channel-wise attention to CNNs.
- ◆ Highlights key regions like defect rings or localized cracks.

◆ Vision Transformers (ViT):

- ◆ Divide image into patches.
- ◆ Use global attention to model long-range defect correlations (e.g., edge + center).

Autoencoders & GANs – Rare Defect Detection and Data Synthesis

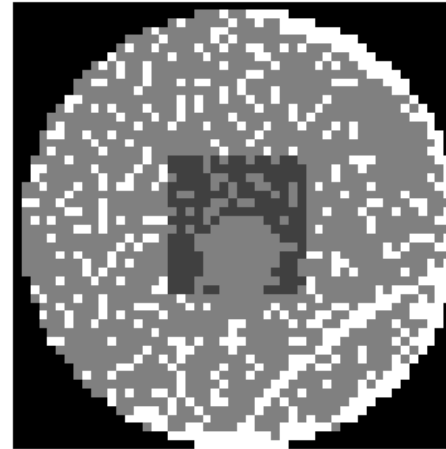
Problem:

- ◆ Some defect types are so rare they have <50 samples.
- ◆ Standard CNNs fail without enough examples.

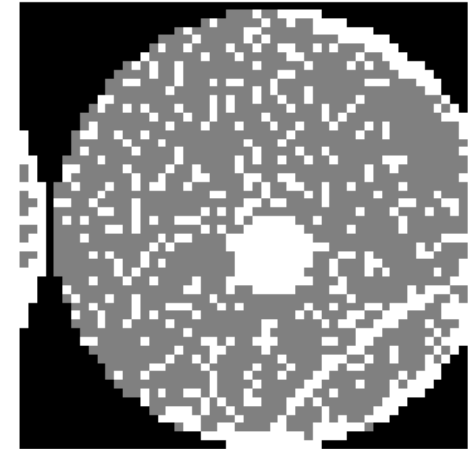
Techniques:

- ◆ **Autoencoders:**
 - Learn to reconstruct normal wafers.
 - High reconstruction error = anomaly.
 - No labels needed.
- ◆ **GANs (Generative Adversarial Networks):**
 - Generate synthetic wafer defects (e.g., ring distortions).
 - Balance skewed datasets.

Autoencoder Output (Simulated Error)



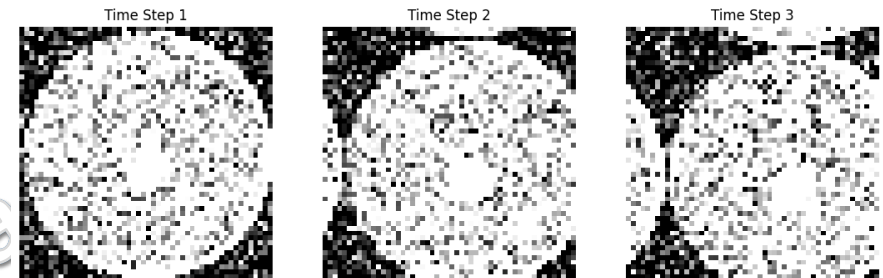
GAN-Generated Variant



Temporal and Hybrid Deep Architectures

📌 Motivation:

- Some defects **don't appear suddenly** — they evolve across:
 - Process steps (etch → deposition → CMP)
 - Time (lot-to-lot drift)



📌 Model Types:

- Temporal CNNs:
 - Use wafer image sequences (or timestamps) as input.
 - Learn how patterns develop / fade over stages.
- CNN + LSTM hybrids:
 - CNN extracts spatial features.
 - LSTM tracks temporal sequence across lots or wafers.
- Spatiotemporal Transformers:
 - Combine both into attention-driven wafer monitoring.





BITS Pilani
Dubai Campus

Thank You!

